

01 Data Setup

The 'Immigration Issue' Misnomer in Human Capital Research: Intersectional Inequalities in Adult Education and Literacy in Upper- and Middle-Income Countries

Blind Peer Review Version

PIAAC PUF Files accessed from <https://www.oecd.org/skills/piaac/data/> in June 2024.

This code is setup up to investigate Cycle 1 for Fig 1, but I use only Cycle 2 for the analysis. The ALE question is not identical and its not a study that probes a longitudinal question.

Setup

```
knitr::opts_chunk$set(warning = FALSE, message = FALSE)

packages <- c('tidyverse',
              'readr',
              'knitr',
              'here',
              'kableExtra',
              'countrycode',
              'readxl')

pacman::p_load(packages, character.only = T)
```

Data

These data were combined from the PUFs of PIAAC using routines built by the author.

They can be found in a companion Github repository

URL Omitted for blind peer review

```
df <- read_csv(here("Data", "piaac_combined.csv"))
df2 <- read_csv(here("Data", "piaac_combined_2.csv"))
```

Recode

Individual Datasets

Cycle 1

```
df <- df %>%
  # subset(PAIDWORK12 == 1 | !C_Q07 %in% c(4, 5, 7, 8, 97) | NFE12JR == 5)
  %>%
  mutate(iso3c = countrycode(as.numeric(CNTRYID), "iso3n", "iso3c"),
         female = case_when(GENDER_R == 2 ~ 1,
                             GENDER_R == 1 ~ 0,
                             TRUE ~ NA_real_),
         male = case_when(GENDER_R == 1 ~ 1,
```

```

        GENDER_R == 2 ~ 0,
        TRUE ~ NA_real_),
age10 = case_when(AGEG10LFS == 1 ~ 20,
        AGEN10LFS == 2 ~ 30,
        AGEN10LFS == 3 ~ 40,
        AGEN10LFS == 4 ~ 50,
        AGEN10LFS == 5 ~ 60,
        TRUE ~ NA_real_
),
# This variable has a disturbing number of missings
FORBORN = ifelse(J_Q04a == "1", 0,
        ifelse(J_Q04a == "2", 1, NA)),
FORBORN_not = (FORBORN-1)*-1,
born_in_country_mother = ifelse(J_Q06a == "1", 1,
        ifelse(J_Q06a == "2", 0, NA)), # includes
guardian if not with biological parents (2nd wave does not)
born_in_country_father = ifelse(J_Q07a == "1", 1,
        ifelse(J_Q07a == "2", 0, NA)),
FORLANG = case_when(BORNLANG == 2 | BORNLANG == 4 ~ 1,
        BORNLANG == 1 | BORNLANG == 3 ~ 0,
        .default = NA),
# fix missing EDLEVEL3 in Austria
EDCAT6 = as.numeric(EDCAT6),
et = case_when(EDCAT6 == 2 | EDCAT6 == 3 ~ 2,
        EDCAT6 == 4 | EDCAT6 == 5 | EDCAT6 == 6 | EDCAT6 ==
7 ~ 3,
        CNTRYID != 40 ~ NA,
        .default = EDCAT6),
ed3 = ifelse(CNTRYID == 40, et, EDLEVEL3),
ed_currently = B_Q02a,
children = case_when(J_Q03a == 2 ~ 0,
        J_Q03a == 1 ~ 1,
        TRUE ~ 0),
#building an equivalent 'num of CVET courses' measure
num1 = as.numeric(B_Q12b), #open or distance education?
num2 = as.numeric(B_Q12d), #on-the-job training
num3 = as.numeric(B_Q12f), #seminars and workshops
num4 = as.numeric(B_Q12h), #courses other not reported
NFE_number = num1 + num2 + num3 + num4,
skill_reading = rowMeans(select(., starts_with("PVLIT")), na.rm = TRUE),
skill_numeracy = rowMeans(select(., starts_with("PVNUM")), na.rm =
TRUE),
group_07 = case_when(children == 1 & female == 1 & FORBORN == 0 ~ 4,
        children == 0 & female == 1 & FORBORN == 0 ~ 3,
        children == 1 & female == 1 & FORBORN == 1 ~ 2,
        children == 0 & female == 1 & FORBORN == 1 ~ 1,
        .default = NA_real_),
literacy_level = case_when( # PIAAC's own 5-category system
        skill_reading < 176 ~ 0,
        skill_reading < 226 ~ 1,
        skill_reading < 276 ~ 2,

```

```

    skill_reading < 326 ~ 3,
    skill_reading > 325 ~ 4,
    TRUE ~ NA_real_
  ),
  numeracy_level = case_when(
    skill_numeracy < 176 ~ 0,
    skill_numeracy < 226 ~ 1,
    skill_numeracy < 276 ~ 2,
    skill_numeracy < 326 ~ 3,
    skill_numeracy > 325 ~ 4,
    TRUE ~ NA_real_
  )
) %>%
select(iso3c, PVLIT1:PVLIT10, PVNUM1:PVNUM10, skill_reading,
skill_numeracy, literacy_level, numeracy_level, NFE12, NFE12JR, female, male,
children, FORLANG, FORBORN, FORBORN_not, BORNLANG, group_07, NFE_number, ed3,
ed_currently, SPFWT0, SPFWT1:SPFWT80) %>%
subset(!is.na(iso3c))

df <- df %>%
  group_by(iso3c) %>%
  # create a weighting variable that is accurate at the country-level, that
  # does not compare populations across countries
  mutate(aweight = SPFWT0 / min(SPFWT0, na.rm = T)) %>%
  ungroup()

# create intersectionalities
df <- df %>%
  mutate(male_not = case_when(male == 1 & FORBORN == 0 ~ 1,
                              male == 0 ~ 0,
                              FORBORN == 1 ~ 0,
                              TRUE ~ NA_real_),
    male_fb = case_when(male == 1 & FORBORN == 1 ~ 1,
                        male == 0 ~ 0,
                        FORBORN == 0 ~ 0,
                        TRUE ~ NA_real_),
    female_not = case_when(female == 1 & FORBORN == 0 ~ 1,
                           female == 0 ~ 0,
                           FORBORN == 1 ~ 0,
                           TRUE ~ NA_real_),
    female_fb = case_when(female == 1 & FORBORN == 1 ~ 1,
                           female == 0 ~ 0,
                           FORBORN == 0 ~ 0,
                           TRUE ~ NA_real_),
    # add Lang
    male_not_natlang = case_when(male == 1 & FORBORN == 0 & FORLANG ==
0 ~ 1,
                                male == 0 ~ 0,
                                FORBORN == 1 ~ 0,
                                FORLANG == 1 ~ 0,
                                TRUE ~ NA_real_),

```

```

male_not_notnatlang = case_when(male == 1 & FORBORN == 0 & FORLANG
== 1 ~ 1,
                                male == 0 ~ 0,
                                FORBORN == 1 ~ 0,
                                FORLANG == 0 ~ 0,
                                TRUE ~ NA_real_),
male_fb_natlang = case_when(male == 1 & FORBORN == 1 & FORLANG == 0
~ 1,
                             male == 0 ~ 0,
                             FORBORN == 0 ~ 0,
                             FORLANG == 1 ~ 0,
                             TRUE ~ NA_real_),
male_fb_notnatlang = case_when(male == 1 & FORBORN == 1 & FORLANG
== 1 ~ 1,
                               male == 0 ~ 0,
                               FORBORN == 0 ~ 0,
                               FORLANG == 0 ~ 0,
                               TRUE ~ NA_real_),
female_not_natlang = case_when(female == 1 & FORBORN == 0 & FORLANG
== 0 ~ 1,
                               female == 0 ~ 0,
                               FORBORN == 1 ~ 0,
                               FORLANG == 1 ~ 0,
                               TRUE ~ NA_real_),
female_not_notnatlang = case_when(female == 1 & FORBORN == 0 &
FORLANG == 1 ~ 1,
                                   female == 0 ~ 0,
                                   FORBORN == 1 ~ 0,
                                   FORLANG == 0 ~ 0,
                                   TRUE ~ NA_real_),
female_fb_natlang = case_when(female == 1 & FORBORN == 1 & FORLANG
== 0 ~ 1,
                              female == 0 ~ 0,
                              FORBORN == 0 ~ 0,
                              FORLANG == 1 ~ 0,
                              TRUE ~ NA_real_),
female_fb_notnatlang = case_when(female == 1 & FORBORN == 1 &
FORLANG == 1 ~ 1,
                                  female == 0 ~ 0,
                                  FORBORN == 0 ~ 0,
                                  FORLANG == 0 ~ 0,
                                  TRUE ~ NA_real_),

# add children
male_not_natlang_child = case_when(male == 1 & FORBORN == 0 &
FORLANG == 0 & children == 1 ~ 1,
                                    male == 0 ~ 0,
                                    FORBORN == 1 ~ 0,
                                    FORLANG == 1 ~ 0,
                                    children == 0 ~ 0,
                                    TRUE ~ NA_real_),
male_not_notnatlang_child = case_when(male == 1 & FORBORN == 0 &

```

```

FORLANG == 1 & children == 1 ~ 1,
    male == 0 ~ 0,
    FORBORN == 1 ~ 0,
    FORLANG == 0 ~ 0,
    children == 0 ~ 0,
    TRUE ~ NA_real_),
    male_fb_natlang_child = case_when(male == 1 & FORBORN == 1 &
FORLANG == 0 & children == 1 ~ 1,
    male == 0 ~ 0,
    FORBORN == 0 ~ 0,
    FORLANG == 1 ~ 0,
    children == 0 ~ 0,
    TRUE ~ NA_real_),
    male_fb_notnatlang_child = case_when(male == 1 & FORBORN == 1 &
FORLANG == 1 & children == 1 ~ 1,
    male == 0 ~ 0,
    FORBORN == 0 ~ 0,
    FORLANG == 0 ~ 0,
    children == 0 ~ 0,
    TRUE ~ NA_real_),
    male_not_natlang_nochild = case_when(male == 1 & FORBORN == 0 &
FORLANG == 0 & children == 0 ~ 1,
    male == 0 ~ 0,
    FORBORN == 1 ~ 0,
    FORLANG == 1 ~ 0,
    children == 1 ~ 0,
    TRUE ~ NA_real_),
    male_not_notnatlang_nochild = case_when(male == 1 & FORBORN == 0 &
FORLANG == 1 & children == 0 ~ 1,
    male == 0 ~ 0,
    FORBORN == 1 ~ 0,
    FORLANG == 0 ~ 0,
    children == 1 ~ 0,
    TRUE ~ NA_real_),
    male_fb_natlang_nochild = case_when(male == 1 & FORBORN == 1 &
FORLANG == 0 & children == 0 ~ 1,
    male == 0 ~ 0,
    FORBORN == 0 ~ 0,
    FORLANG == 1 ~ 0,
    children == 1 ~ 0,
    TRUE ~ NA_real_),
    male_fb_notnatlang_nochild = case_when(male == 1 & FORBORN == 1 &
FORLANG == 1 & children == 0 ~ 1,
    male == 0 ~ 0,
    FORBORN == 0 ~ 0,
    FORLANG == 0 ~ 0,
    children == 1 ~ 0,
    TRUE ~ NA_real_),
    female_not_natlang_child = case_when(female == 1 & FORBORN == 0 &
FORLANG == 0 & children == 1 ~ 1,
    female == 0 ~ 0,

```

```

FORBORN == 1 ~ 0,
FORLANG == 1 ~ 0,
children == 0 ~ 0,
TRUE ~ NA_real_),
female_not_notnatlang_child = case_when(female == 1 & FORBORN == 0
& FORLANG == 1 & children == 1 ~ 1,
female == 0 ~ 0,
FORBORN == 1 ~ 0,
FORLANG == 0 ~ 0,
children == 0 ~ 0,
TRUE ~ NA_real_),
female_fb_natlang_child = case_when(female == 1 & FORBORN == 1 &
FORLANG == 0 & children == 1 ~ 1,
female == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 1 ~ 0,
children == 0 ~ 0,
TRUE ~ NA_real_),
female_fb_notnatlang_child = case_when(female == 1 & FORBORN == 1 &
FORLANG == 1 & children == 1 ~ 1,
female == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 0 ~ 0,
children == 0 ~ 0,
TRUE ~ NA_real_),
female_not_natlang_nochild = case_when(female == 1 & FORBORN == 0 &
FORLANG == 0 & children == 0 ~ 1,
female == 0 ~ 0,
FORBORN == 1 ~ 0,
FORLANG == 1 ~ 0,
children == 1 ~ 0,
TRUE ~ NA_real_),
female_not_notnatlang_nochild = case_when(female == 1 & FORBORN ==
0 & FORLANG == 1 & children == 0 ~ 1,
female == 0 ~ 0,
FORBORN == 1 ~ 0,
FORLANG == 0 ~ 0,
children == 1 ~ 0,
TRUE ~ NA_real_),
female_fb_natlang_nochild = case_when(female == 1 & FORBORN == 1 &
FORLANG == 0 & children == 0 ~ 1,
female == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 1 ~ 0,
children == 1 ~ 0,
TRUE ~ NA_real_),
female_fb_notnatlang_nochild = case_when(female == 1 & FORBORN == 1
& FORLANG == 1 & children == 0 ~ 1,
female == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 0 ~ 0,

```

```

        children == 1 ~ 0,
        TRUE ~ NA_real_),
    )

```

Cycle 2

B2_Q05a - **Are you currently studying for any kind of formal qualification?** c(Yes = 1, No = 2)

B2_Q22 - **In the last 12 months, were there any training activities you wanted to participate in but did not?** c(Yes = 1, No = 2)

B2_Q23 - **Which of the following reasons prevented you from participating in these training activities? Please indicate the most important reason.** c('I did not find any suitable training activity' = 1, 'I had no time due to family responsibilities' = 2, 'I had no time due to work-related reasons' = 3, 'I did not have the prerequisites' = 4, 'Lack of employer's support' = 5, 'Training activity would have been too expensive' = 6, 'Training activity took place at an inconvenient time or location' = 7, 'Training activity was canceled or postponed' = 8, 'Something unexpected came up that prevented me from participating' = 9, 'Other reason' = 10)

H2_Q19a - **Overall, which of the following statements best describes your skills in relation to what is required to do your job?** c('My skills are higher than required by my job' = 1, 'My skills are matched to what is required by my job' = 2, 'Some of my skills are lower than what is required by my job and need to be further developed' = 3)

```

df2 <- df2 %>%
  mutate(
    iso3c = countrycode(as.numeric(CNTRYID), "iso3n", "iso3c"),
    iso3c = ifelse(as.numeric(CNTRYID_E) == 7561, "CHE",
                  ifelse(as.numeric(CNTRYID_E) == 7562, "CHE",
                        ifelse(as.numeric(CNTRYID_E) == 7563, "CHE",
                              ifelse(as.numeric(CNTRYID_E) == 152, "CHL",
                                    ifelse(as.numeric(CNTRYID_E) == 203, "CZE", iso3c)))))
    skill_reading = rowMeans(select(., starts_with("PVLIT")), na.rm = TRUE),
    skill_numeracy = rowMeans(select(., starts_with("PVNUM")), na.rm = TRUE),
    literacy_level = case_when( # PIAAC's own 5-category system
      skill_reading < 176 ~ 0,
      skill_reading < 226 ~ 1,
      skill_reading < 276 ~ 2,
      skill_reading < 326 ~ 3,
      skill_reading > 325 ~ 4,
      TRUE ~ NA_real_
    ),
    numeracy_level = case_when(
      skill_numeracy < 176 ~ 0,
      skill_numeracy < 226 ~ 1,
      skill_numeracy < 276 ~ 2,
      skill_numeracy < 326 ~ 3,
      skill_numeracy > 325 ~ 4,

```

```

    TRUE ~ NA_real_
  ),
  FORBORN = case_when(BORNLANG == "1" | BORNLANG == "2" ~ 0,
                      BORNLANG == "3" | BORNLANG == "4" ~ 1,
                      .default = NA),
  # Denmark missing values
  FORBORN = case_when(CNTRYID == 208 & A2_Q03a == "2" ~ 1,
                      CNTRYID == 208 & A2_Q03a == "1" ~ 0,
                      .default = FORBORN),
  FORBORN_not = (FORBORN-1)*-1,
  FORLANG = case_when(BORNLANG == 1 | BORNLANG == 3 ~ 0,
                      BORNLANG == 2 | BORNLANG == 4 ~ 1,
                      .default = NA),
  EDCAT6 = as.numeric(EDCAT6_TC1),
  EDCAT7 = as.numeric(EDCAT7_TC1),
  ed3 = case_when(EDCAT6 == 2 | EDCAT6 == 3 ~ 2,
                  EDCAT6 == 4 | EDCAT6 == 5 | EDCAT6 == 6 | EDCAT6 == 7 ~
3,
                  .default = EDCAT6),
  ed3_backup = case_when(EDCAT7 == 1 ~ 1,
                        EDCAT7 > 1 & EDCAT7 < 5 ~ 2,
                        EDCAT7 > 4 ~ 3,
                        TRUE ~ NA_real_),
  ed3 = ifelse(is.na(ed3), ed3_backup, ed3),
  ed_currently = ifelse(B2_Q05a == "1", 1, 0),
  ed_currently = ifelse(is.na(ed_currently), 0, ed_currently), # recode
missing to 0 for parsimony
  NFE_barrier = ifelse(B2_Q22 == "1", 1, 0),
  NFE_barrier = ifelse(is.na(NFE_barrier), 0, NFE_barrier), # recode
missing to 0 for parsimony
  underskilled = case_when(H2_Q19a == "3" ~ 1,
                          .default = 0),

  year = 2022,
  A2_N02_T = as.numeric(A2_N02_T),
  female = ifelse(A2_N02_T == 2, 1, 0),
  female_2 = ifelse(GENDER_R == 2, 1, 0),
  female = ifelse(is.na(female), female_2, female),
  male = case_when(female == 0 ~ 1,
                  female == 1 ~ 0,
                  TRUE ~ NA_real_),
  children = ifelse(J2_Q03a == 1, 1, 0),
  children = ifelse(is.na(children), 0, children),
  group_07 = case_when(children == 1 & female == 1 & FORBORN == 0 ~ 4,
                      children == 0 & female == 1 & FORBORN == 0 ~ 3,
                      children == 1 & female == 1 & FORBORN == 1 ~ 2,
                      children == 0 & female == 1 & FORBORN == 1 ~ 1,
                      .default = NA_real_),
  NFE_number = as.numeric(NFE12NR),
  NFE12JR = NFE12JRC2,
  NFE12 = ifelse(B2_Q08a == "2", "0",
                ifelse(B2_Q08a == "1", "1", NA)), # participated in any training

```

Last 12 months

```
NFE12 = ifelse(is.na(NFE12), NFE12C2, NFE12),  
NFE12 = as.numeric(NFE12),  
NFE12JRa = as.numeric(NFE12JRC2), # note that this variable is coded  
differently than in wave one. This wave, those who did not participate in any  
training are coded as missing, so that this variable is only the percentage  
of those who reported training that reported it was job-related. This has to  
be fixed to compare them
```

```
NFE12JR = ifelse(!is.na(NFE12) & is.na(NFE12JRa), "0",
as.character(NFE12JRa))) %>%
  select(iso3c, PVLIT1:PVLIT10, PVNUM1:PVNUM10, skill_reading,
skill_numeracy, literacy_level, numeracy_level, NFE12, NFE12JR, NFE_barrier,
underskilled, female, male, children, FORLANG, FORBORN, FORBORN_not,
BORNLANG, group_07, NFE_number, ed3, ed_currently, SPFWT0, SPFWT1:SPFWT80)
```

```
df2 <- df2 %>%
  group_by(iso3c) %>%
  # create a weighting variable that is accurate at the country-level, that
  # does not compare populations across countries
  mutate(aweight = SPFWT0 / min(SPFWT0, na.rm = T)) %>%
  ungroup()
```

```
# create intersectionalities
```

```
df2 <- df2 %>%
  mutate(male_not = case_when(male == 1 & FORBORN == 0 ~ 1,
                              male == 0 ~ 0,
                              FORBORN == 1 ~ 0,
                              TRUE ~ NA_real_),
         male_fb = case_when(male == 1 & FORBORN == 1 ~ 1,
                              male == 0 ~ 0,
                              FORBORN == 0 ~ 0,
                              TRUE ~ NA_real_),
         female_not = case_when(female == 1 & FORBORN == 0 ~ 1,
                                female == 0 ~ 0,
                                FORBORN == 1 ~ 0,
                                TRUE ~ NA_real_),
         female_fb = case_when(female == 1 & FORBORN == 1 ~ 1,
                                female == 0 ~ 0,
                                FORBORN == 0 ~ 0,
                                TRUE ~ NA_real_),
         # add Lang
         male_not_natlang = case_when(male == 1 & FORBORN == 0 & FORLANG ==
0 ~ 1,
                                     male == 0 ~ 0,
                                     FORBORN == 1 ~ 0,
                                     FORLANG == 1 ~ 0,
                                     TRUE ~ NA_real_),
         male_not_notnatlang = case_when(male == 1 & FORBORN == 0 & FORLANG
== 1 ~ 1,
                                     male == 0 ~ 0,
                                     FORBORN == 1 ~ 0,
```

```

FORLANG == 0 ~ 0,
TRUE ~ NA_real_),
male_fb_natlang = case_when(male == 1 & FORBORN == 1 & FORLANG == 0
~ 1,
male == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 1 ~ 0,
TRUE ~ NA_real_),
male_fb_notnatlang = case_when(male == 1 & FORBORN == 1 & FORLANG
== 1 ~ 1,
male == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 0 ~ 0,
TRUE ~ NA_real_),
female_not_natlang = case_when(female == 1 & FORBORN == 0 & FORLANG
== 0 ~ 1,
female == 0 ~ 0,
FORBORN == 1 ~ 0,
FORLANG == 1 ~ 0,
TRUE ~ NA_real_),
female_not_notnatlang = case_when(female == 1 & FORBORN == 0 &
FORLANG == 1 ~ 1,
female == 0 ~ 0,
FORBORN == 1 ~ 0,
FORLANG == 0 ~ 0,
TRUE ~ NA_real_),
female_fb_natlang = case_when(female == 1 & FORBORN == 1 & FORLANG
== 0 ~ 1,
female == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 1 ~ 0,
TRUE ~ NA_real_),
female_fb_notnatlang = case_when(female == 1 & FORBORN == 1 &
FORLANG == 1 ~ 1,
female == 0 ~ 0,
FORBORN == 0 ~ 0,
FORLANG == 0 ~ 0,
TRUE ~ NA_real_),

# add children
male_not_natlang_child = case_when(male == 1 & FORBORN == 0 &
FORLANG == 0 & children == 1 ~ 1,
male == 0 ~ 0,
FORBORN == 1 ~ 0,
FORLANG == 1 ~ 0,
children == 0 ~ 0,
TRUE ~ NA_real_),
male_not_notnatlang_child = case_when(male == 1 & FORBORN == 0 &
FORLANG == 1 & children == 1 ~ 1,
male == 0 ~ 0,
FORBORN == 1 ~ 0,
FORLANG == 0 ~ 0,

```

```

        children == 0 ~ 0,
        TRUE ~ NA_real_),
    male_fb_natlang_child = case_when(male == 1 & FORBORN == 1 &
FORLANG == 0 & children == 1 ~ 1,
        male == 0 ~ 0,
        FORBORN == 0 ~ 0,
        FORLANG == 1 ~ 0,
        children == 0 ~ 0,
        TRUE ~ NA_real_),
    male_fb_notnatlang_child = case_when(male == 1 & FORBORN == 1 &
FORLANG == 1 & children == 1 ~ 1,
        male == 0 ~ 0,
        FORBORN == 0 ~ 0,
        FORLANG == 0 ~ 0,
        children == 0 ~ 0,
        TRUE ~ NA_real_),
    male_not_natlang_nochild = case_when(male == 1 & FORBORN == 0 &
FORLANG == 0 & children == 0 ~ 1,
        male == 0 ~ 0,
        FORBORN == 1 ~ 0,
        FORLANG == 1 ~ 0,
        children == 1 ~ 0,
        TRUE ~ NA_real_),
    male_not_notnatlang_nochild = case_when(male == 1 & FORBORN == 0 &
FORLANG == 1 & children == 0 ~ 1,
        male == 0 ~ 0,
        FORBORN == 1 ~ 0,
        FORLANG == 0 ~ 0,
        children == 1 ~ 0,
        TRUE ~ NA_real_),
    male_fb_natlang_nochild = case_when(male == 1 & FORBORN == 1 &
FORLANG == 0 & children == 0 ~ 1,
        male == 0 ~ 0,
        FORBORN == 0 ~ 0,
        FORLANG == 1 ~ 0,
        children == 1 ~ 0,
        TRUE ~ NA_real_),
    male_fb_notnatlang_nochild = case_when(male == 1 & FORBORN == 1 &
FORLANG == 1 & children == 0 ~ 1,
        male == 0 ~ 0,
        FORBORN == 0 ~ 0,
        FORLANG == 0 ~ 0,
        children == 1 ~ 0,
        TRUE ~ NA_real_),
    female_not_natlang_child = case_when(female == 1 & FORBORN == 0 &
FORLANG == 0 & children == 1 ~ 1,
        female == 0 ~ 0,
        FORBORN == 1 ~ 0,
        FORLANG == 1 ~ 0,
        children == 0 ~ 0,
        TRUE ~ NA_real_),

```

```

    female_not_notnatlang_child = case_when(female == 1 & FORBORN == 0
& FORLANG == 1 & children == 1 ~ 1,
      female == 0 ~ 0,
      FORBORN == 1 ~ 0,
      FORLANG == 0 ~ 0,
      children == 0 ~ 0,
      TRUE ~ NA_real_),
    female_fb_natlang_child = case_when(female == 1 & FORBORN == 1 &
FORLANG == 0 & children == 1 ~ 1,
      female == 0 ~ 0,
      FORBORN == 0 ~ 0,
      FORLANG == 1 ~ 0,
      children == 0 ~ 0,
      TRUE ~ NA_real_),
    female_fb_notnatlang_child = case_when(female == 1 & FORBORN == 1 &
FORLANG == 1 & children == 1 ~ 1,
      female == 0 ~ 0,
      FORBORN == 0 ~ 0,
      FORLANG == 0 ~ 0,
      children == 0 ~ 0,
      TRUE ~ NA_real_),
    female_not_natlang_nochild = case_when(female == 1 & FORBORN == 0 &
FORLANG == 0 & children == 0 ~ 1,
      female == 0 ~ 0,
      FORBORN == 1 ~ 0,
      FORLANG == 1 ~ 0,
      children == 1 ~ 0,
      TRUE ~ NA_real_),
    female_not_notnatlang_nochild = case_when(female == 1 & FORBORN ==
0 & FORLANG == 1 & children == 0 ~ 1,
      female == 0 ~ 0,
      FORBORN == 1 ~ 0,
      FORLANG == 0 ~ 0,
      children == 1 ~ 0,
      TRUE ~ NA_real_),
    female_fb_natlang_nochild = case_when(female == 1 & FORBORN == 1 &
FORLANG == 0 & children == 0 ~ 1,
      female == 0 ~ 0,
      FORBORN == 0 ~ 0,
      FORLANG == 1 ~ 0,
      children == 1 ~ 0,
      TRUE ~ NA_real_),
    female_fb_notnatlang_nochild = case_when(female == 1 & FORBORN == 1
& FORLANG == 1 & children == 0 ~ 1,
      female == 0 ~ 0,
      FORBORN == 0 ~ 0,
      FORLANG == 0 ~ 0,
      children == 1 ~ 0,
      TRUE ~ NA_real_),
  )

```

Country Indicators

GDP

<https://dataverse.nl/dataset.xhtml?persistentId=doi:10.34894/INZBF2>

```
#get all countries
clist <- unique(df2$iso3c)

df_gdp <- read_excel(here("Data", "mpd2023_web.xlsx"), sheet = "Full data",
  col_types = c("text", "text", "text", "numeric", "numeric", "numeric")) %>%
  mutate(iso3c = countrycode) %>%
  subset(iso3c %in% clist) %>%
  subset(year > 2018) %>%
  select(iso3c, gdppc, pop) %>%
  group_by(iso3c) %>%
  summarise_all(mean, na.rm = T) %>%
  ungroup()
```

Save

```
df2 <- df2 %>%
  left_join(df_gdp, by = c("iso3c")) %>%
  mutate(gdppc_cap = ifelse(gdppc < 60000, gdppc, 60000),
    gdppc_k = gdppc_cap/1000,
    gdppc_k_log = log(gdppc_k))

write_rds(df, here("Data", "df.RDS"))

write_rds(df2, here("Data", "df2.RDS"))
```

Citations

```
packages %>%
  map(citation) %>%
  print(style = "text")

## [[1]]
## Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R,
## Grolemund G, Hayes A, Henry L, Hester J, Kuhn M, Pedersen TL, Miller E,
## Bache SM, Müller K, Ooms J, Robinson D, Seidel DP, Spinu V, Takahashi
## K, Vaughan D, Wilke C, Woo K, Yutani H (2019). "Welcome to the
## tidyverse." _Journal of Open Source Software_, 4(43), 1686.
## doi:10.21105/joss.01686 <https://doi.org/10.21105/joss.01686>.
##
## [[2]]
## Wickham H, Hester J, Bryan J (2024). _readr: Read Rectangular Text
## Data_. R package version 2.1.5,
## <https://CRAN.R-project.org/package=readr>.
##
## [[3]]
```

```
## Xie Y (2025). _knitr: A General-Purpose Package for Dynamic Report
## Generation in R_. R package version 1.50, <https://yihui.org/knitr/>.
##
## Xie Y (2015). _Dynamic Documents with R and knitr_, 2nd edition.
## Chapman and Hall/CRC, Boca Raton, Florida. ISBN 978-1498716963,
## <https://yihui.org/knitr/>.
##
## Xie Y (2014). "knitr: A Comprehensive Tool for Reproducible Research in
## R." In Stodden V, Leisch F, Peng RD (eds.), _Implementing Reproducible
## Computational Research_. Chapman and Hall/CRC. ISBN 978-1466561595.
##
## [[4]]
## Müller K (2020). _here: A Simpler Way to Find Your Files_. R package
## version 1.0.1, <https://CRAN.R-project.org/package=here>.
##
## [[5]]
## Zhu H (2024). _kableExtra: Construct Complex Table with 'kable' and
## Pipe Syntax_. R package version 1.4.0,
## <https://CRAN.R-project.org/package=kableExtra>.
##
## [[6]]
## Arel-Bundock V, Enevoldsen N, Yetman C (2018). "countrycode: An R
## package to convert country names and country codes." _Journal of Open
## Source Software_, *3*(28), 848. <https://doi.org/10.21105/joss.00848>.
##
## [[7]]
## Wickham H, Bryan J (2025). _readxl: Read Excel Files_. R package
## version 1.4.5, <https://CRAN.R-project.org/package=readxl>.
```

Colophon

sessionInfo()

```
## R version 4.4.1 (2024-06-14 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
```

```
##
## other attached packages:
## [1] readxl_1.4.5      countrycode_1.6.0 kableExtra_1.4.0 here_1.0.1
## [5] knitr_1.50        lubridate_1.9.4   forcats_1.0.0   stringr_1.5.1
## [9] dplyr_1.1.4       purrr_1.0.4       readr_2.1.5     tidyr_1.3.1
## [13] tibble_3.2.1      ggplot2_3.5.1     tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] generics_0.1.3    xml2_1.3.8        stringi_1.8.4     hms_1.1.3
## [5] digest_0.6.37     magrittr_2.0.3    evaluate_1.0.3    grid_4.4.1
## [9] timechange_0.3.0  fastmap_1.2.0     cellranger_1.1.0  rprojroot_2.0.4
## [13] viridisLite_0.4.2 scales_1.3.0       cli_3.6.4         crayon_1.5.3
## [17] rlang_1.1.5       bit64_4.6.0-1     munsell_0.5.1     withr_3.0.2
## [21] yaml_2.3.10       parallel_4.4.1    tools_4.4.1       tzdb_0.5.0
## [25] colorspace_2.1-1  pacman_0.5.1      vctrs_0.6.5       R6_2.6.1
## [29] lifecycle_1.0.4   bit_4.6.0         vroom_1.6.5       pkgconfig_2.0.3
## [33] pillar_1.10.1     gtable_0.3.6      glue_1.8.0
## [37] xfun_0.51         tidyselect_1.2.1  rstudioapi_0.17.1
## [41] rmarkdown_2.29    svglite_2.1.3     compiler_4.4.1
systemfonts_1.2.1
htmltools_0.5.8.1
```